Facultad de Defensa Nacional

Carrera Lic. En Ciberdefensa

Análisis Matemático I

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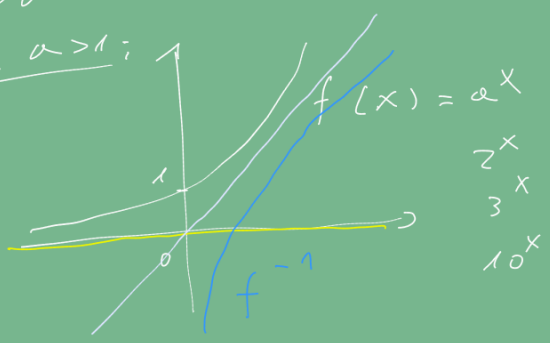
Año: 2025

Carga horaria: 64

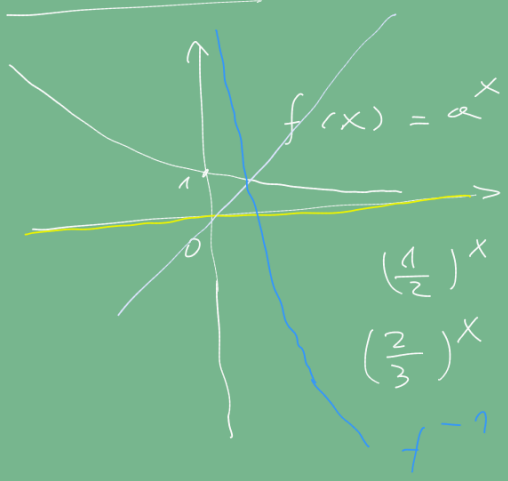
LOGARITMOS:

FUNCIÓN EXPONENCIAL: $f: \mathbb{R} \longrightarrow (0,+\infty)$, $f(x) = a^x$, DONDE $0 < a \neq 1$.

SI $a > 1$:



SI $0 < a < 1$:



$f^{-1}: (0,+\infty) \longrightarrow \mathbb{R}$

$f^{-1} = \log_a$ (LOGARITMO EN BASE a).

PARA $y > 0$, $x \in \mathbb{R}$: $\log_a(y) = x \iff a^x = y$

$(0 < a \neq 1)$

EJ:

1) $\log_a(1) = 0$

2) $\log_3(9) = 2$

3) $\log_4\left(\frac{1}{8}\right) = ?$ $\quad \frac{1}{8} = \frac{1}{2^3} = \frac{1}{(\sqrt{4})^3} = \frac{1}{(4^{\frac{1}{2}})^3} = \frac{1}{4^{3/2}} =$

$= 4^{-\frac{3}{2}} \Rightarrow \log_4\left(\frac{1}{8}\right) = -\frac{3}{2}$

4) $\log_a(a^r) = r$

5) $\log_a(a) = 1$

PROPIEDADES:

SEAN $0 < a \neq 1$, $X > 0$, $Y > 0$, $r \in \mathbb{R}$.

1) $\log_a(X \cdot Y) = \log_a(X) + \log_a(Y)$

2) $\log_a\left(\frac{X}{Y}\right) = \log_a(X) - \log_a(Y)$

3) $\log_a(X^r) = r \log_a(X)$

OBS: $\log_a\left(\frac{-5}{-3}\right) = \log_a(-5) - \log_a(-3)$

$\frac{-5}{-3} > 0$

↘ NO EXISTEN ↙

SÍ: $\log_a\left(\frac{5}{3}\right) = \log_a(5) - \log_a(3)$.

BASES USUALES:

$\log_{10}(X) = \log(X)$ (LOGARITMO DECIMAL)

$\log_e(X) = \ln(X)$ (LOGARITMO NATURAL).

$e = 2{,}7181828\ldots$ → NÚMERO e

CAMBIO DE BASE:

$$\log_a(X) = \frac{\log_c(X)}{\log_c(a)}$$

EJ: $\log_5(27) = \frac{\log(27)}{\log(5)} = \frac{\ln(27)}{\ln(5)}$

OBS: (1) $a^{\log_a(X)} = X \quad (\forall X > 0)$

(2) $\log_a(a^X) = X \quad (\forall X \in \mathbb{R})$

OBS: SI $0<a\neq 1$, $x>0$, $y>0$, ENTONCES

$$\log_a(x\cdot y) = \log_a(x) + \log_a(y)$$

$$a^{\log_a(x)+\log_a(y)} = a^{\log_a(x)}\cdot a^{\log_a(y)} = x\cdot y \checkmark$$

OBS: SI $0<a\neq 1$, $r\in\mathbb{R}$, $x>0$, ENTONCES

$$\log_a(x^r) = r\log_a(x)$$

$$a^{r\log_a(x)} = \left(a^{\log_a(x)}\right)^r = x^r \checkmark$$

CAMBIO DE BASE:

$$\log_a(x) = \frac{\log_c(x)}{\log_c(a)}$$

D/

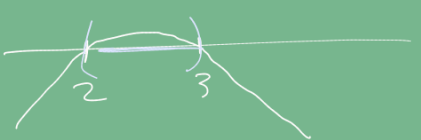
$$a^{\frac{\log_c(x)}{\log_c(a)}} = \left(c^{\log_c(a)}\right)^{\frac{\log_c(x)}{\log_c(a)}} = c^{\log_c(x)} = x$$

EJ: HALLAR EL DOMINIO DE $f(x) = \dfrac{\ln(-x^2+5x-6)}{x}$.

S/

$$-x^2+5x-6 = -(x^2-5x+6) = -(x-2)\cdot(x-3)$$

$$\Rightarrow f(x) = \frac{\ln\left(\overset{>0}{-(x-2)\cdot(x-3)}\right)}{x\neq 0}$$



$$\therefore \boxed{D_{\max}(f) = (2,3)}.$$

Ej.: $f(x) = \dfrac{\ln\left((-x^2+5x-6)^{\circ}\right)}{2x-5}$

$2x-5 \neq 0 \iff x \neq \frac{5}{2} = 2{,}5$

$\therefore \boxed{D_{\text{m}}(f) = (2, \tfrac{5}{2}) \cup (\tfrac{5}{2}, 3)}$

